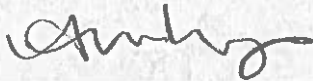


DSC 102 – Spring 2026 – Midterm Exam

May 7th 2026

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By signing below, you agree that you will behave honestly and fairly throughout and after this exam. You should not discuss any part of this exam with anyone who has not yet taken it.

Signature	
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Instructions:

- You have 60 minutes to complete the exam.
- Your exam version should be different from those to your person left/right. If not, notify the instructor.
- Write your answers in the dedicated box spaces as directed, and ensure your writing is clear and legible.
- Write your name or PID at the top of each sheet in the space provided.
- No calculators are permitted.
- Up to 2 letter sheets (4 pages) of any notes/material are allowed. Use the blank pages at the end of the exam to make scratch notes/calculations. The exam is otherwise closed: No electronics, books, other notes, or communication with peers allowed.

(Please wait until you are instructed to open and start your exam.)

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Number Representations

Question 1

[1+2 points] An autonomous vehicle's event-logging system assigns a unique unsigned binary code to each type of sensor event. The engineering team has defined 245 distinct event types in the current firmware release.

(a) What is the minimum number of bits required to represent all 245 event types?

Answer:

8

$$\begin{array}{r} 128 \\ 127 \\ \hline 255 \end{array}$$

(b) Using that bit-width, event type 198 (labeled "lane-departure warning") is stored as the binary value:

Answer:

1100 0110

128 64 32 16 8 4 2 1

$$1 + 2 + 4 + 8 + 128 = 143$$

$$\begin{array}{r} 128 \\ 128 \\ \hline 256 \end{array}$$

$$\begin{array}{r} 128 \\ 128 \\ \hline 256 \end{array}$$

$$\begin{array}{r} 128 \\ 128 \\ \hline 256 \end{array}$$

Question 2

[2 points] A network monitoring tool captures raw 8-bit bytes from packet headers (each in the range 0–255). A security analyst finds that a suspicious header field holds the decimal value 204. Express this byte as a 2-digit hexadecimal value:

Answer: 0x

CC

Question 3

[2+1 points] A robotics control system stores joint-angle gradient updates and must choose between two 16-bit floating-point formats:

– float16: 1 sign / 5 exponent / 10 mantissa bits

– bfloat16: 1 sign / 8 exponent / 7 mantissa bits

(a) Following the IEEE-754 convention, $e = E - \text{bias}$. The bias for a 5-bit exponent in float16 should be:

Answer:

15

$$2^{5-1} - 1 = 2^4 - 1 = 16 - 1 = 15$$

(b) When small differences in joint-angle corrections must be preserved (i.e., fine-grained precision matters more than range), the format offering higher precision is:

Answer:

float16

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$$\begin{array}{r} 1 \\ 32 \\ 16 \\ 14 \\ \hline 62 \end{array}$$

Question 4

[2 points] A smart building's elevator controller logs floor-position deltas (the signed difference between the requested floor and the current floor) as 8-bit signed integers in two's complement. Encode the delta -61 floors as an 8-bit two's complement bit pattern:

Answer:

1100 0011

$$\begin{array}{r} 64 \\ 3 \\ \hline 67 \end{array} \quad \begin{array}{r} 128 \\ 67 \\ \hline 61 \end{array}$$

$$\begin{array}{r} -128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 61 = 0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \end{array}$$

$$\begin{array}{r} 11 \ 00 \ 00 \ 10 \\ \downarrow \end{array}$$

$$\begin{array}{r} 11 \ 00 \ 00 \ 11 \end{array}$$

$$\begin{array}{r} 32 \ 56 \ 11 \\ 32 \\ \hline 29 \end{array} \quad \begin{array}{r} 16 \ 29 \\ 16 \\ \hline 13 \end{array} \quad \begin{array}{r} 12 \ 13 \\ 12 \\ \hline 1 \end{array} \quad \begin{array}{r} 11 \\ 13 \\ \hline 8 \end{array}$$

$$\begin{array}{r} 24 \ 21 \end{array}$$

Question 5

[2 points] A hospital patient-intake system uses a signed 8-bit integer counter (range -128 to 127) to tally admissions from two intake wings. Each wing admits exactly 110 patients, so the system expects a total of 220. Due to integer overflow in two's-complement arithmetic, the counter actually displays the decimal value:

Answer:

-36

$$\begin{array}{r} -128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0 \end{array}$$

$$\begin{array}{r} 2 \\ 128 \\ 64 \\ 16 \\ 12 \\ \hline 220 \checkmark \end{array} \quad \begin{array}{r} 1 \\ 64 \\ 16 \\ 12 \\ \hline 92 \end{array} \quad \begin{array}{r} -128 \\ +92 \\ \hline -36 \end{array}$$

$$\begin{array}{r} 128 \ 220 \ 10 \\ 21 \ 128 \\ \hline 92 \end{array} \quad \begin{array}{r} 128 \\ 64 \ 12 \\ 8 \ 64 \\ \hline 28 \end{array} \quad \begin{array}{r} 128 \\ 16 \\ \hline 12 \end{array} \quad \begin{array}{r} 124 \\ 8 \ 12 \\ \hline 8 \\ 4 \end{array}$$

Computer Organization / Architecture

Question 1:

[2+2+2 points] A processor on a data server runs at speed $PS = 5 \text{ GHz}$. A program has following instruction mix:

- Arithmetic instructions $AR = 60\%$, each taking $nAR = 1$ cycle to execute.
- Memory instructions $ME = 20\%$, each taking $nME = 5$ cycles to execute.
- Control instructions $CO = 20\%$, each taking $nCO = 2$ cycles to execute.

Total instructions executed $TI = 5 \times 10^9$

1. What is the average CPI (Clock cycles per instruction)?

Answer: 2 CPI

$$0.6(1) + 0.2(5) + 0.2(2)$$

$$0.6 + 1 + 0.4 = 2$$

(5B)

$$5 \times 10^9 * 2 = 10 \times 10^9$$

2. What is the total number of cycles TC ?

Answer: 1×10^{10} cycles

3. What is the execution time ET ?

$$\frac{1 \times 10^{10}}{5 \times 10^9} = \frac{2(5 \times 10^9)}{5 \times 10^9} = 2$$

Answer: 2 second(s)

Question 2:

[1+1+2 points] You want to optimize a program that you write for a customer by taking advantage of CPU cache on the customer computer. Here is the related information you have about your code and your customer hardware:

- Your program has a total of $TI = 3$ billion instructions.
- Memory operations are $ME = 40\%$ of the TI . Each takes $nME = 1$ cycle plus a stall of $nST = 60$ cycles to execute.
- The CPU clock is $PS = 3 \text{ GHz}$.
- Non-memory instructions take 1 cycle.

1. How many cycles are spent on base instruction execution?

Answer: 4.3×10^{10} cycles

$$0.4(1) + 0.6(1) = 1 \text{ CPI}$$

$$TC = 3 \times 10^9 (25) =$$

$$1 + 0.4(60) = 1 + 24 = 25 \text{ CPI}$$

$$7.5 \times 10^{10} \text{ cycles}$$

$$\#SC = (3 \times 10^9 * 0.4 * 60)$$

$$3B \times 24 = 32B \text{ stall cycles}$$

75B

2. How many cycles are spent waiting on memory?

Answer: 3.2×10^{10} cycles

$$ET = TC / 3 \times 10^4$$

$$75B / 3B = 25B$$

3. What is the total execution time ET?

Answer: 25 second(s)

Question 3

[2+2 points] A bioinformatics team is evaluating a DNA sequence-alignment kernel running on a 5-stage pipelined CPU. Profiling reveals the following stage latencies:

Stage	IF	ID	EX	MEM	WB
Latency (ns)	0.22	0.18	0.45	0.28	0.12

A ns = 10^{-9} seconds.

The alignment kernel executes 4×10^9 instructions on this ideally pipelined processor (no stalls; one instruction completes every clock cycle in steady state).

a. What is the maximum allowable clock frequency (in GHz) for this design?

Answer: 4

b. At that clock rate, what is the total wall-clock time (in seconds) to complete the full 4×10^9 instruction workload? Ignore pipeline fill time.

Answer: 2

Operating Systems (CPU Virtualization, Memory Virtualization, Persistence)

Question 1

[2+2+2 points] A logistics company runs a fleet-tracking service as **5 isolated vehicle-agent processes** on a low-power embedded controller. The memory system uses:

- A 15-bit virtual address space per process 2^{15} bytes
 - 64-byte pages 2^6 bytes
 - 4-byte page table entries (PTEs) 4 bytes
- $\text{page size} = 64 \text{ bytes}$
 $2^k \Rightarrow k = 6$

(a) The 15-bit virtual address is split into a virtual page number (VPN) and a page offset. How many bits are allocated to each field?

Answer: Page offset = 6 bits, VPN = 9 bits

(b) What is the total memory consumed by the page tables of all 5 processes combined?

Answer: 2" bytes

$\# \text{pages} * \text{size of each entry}$
 $2^9 * 4 \text{ bytes}$

(c) A vehicle agent issues a load to virtual address **0x5E3A**. The PTE for the corresponding VPN reports physical frame number (PFN) = **0x17**. Compute the physical address.

Answer: 0x 173A

$$\# \text{pages} = 2^{15} / 64 \text{ byte} = \frac{2^{15}}{2^6} = 2^9 \text{ pages}$$

$$2^9 * 2^2 \text{ bytes} = 2^{11} \text{ bytes} \Rightarrow$$

94(4)

$$\begin{array}{r} 5E \\ 5 \quad 14 \\ 0101 \quad 1110 \\ 64 \quad 16 \quad 842 \\ 1 \\ 64 \\ 16 \\ 14 \\ \hline 94 \end{array}$$

$$\begin{array}{r} 17 \\ 0001 \quad 0111 \\ 16 \quad 421 \\ 16 \\ 7 \\ \hline 23(4) \end{array}$$

Question 2

[3+2 points] A data engineering team launches a batch ETL pipeline using Python's process API. Consider the following program (assume every `os.fork()` call succeeds and no process exits before the MARK point):

```
import os
os.fork()                # line A
if os.fork() == 0:      # line B
    os.fork()           # line C - child of line B only
    os.fork()           # line D - child of line B only
os.fork()               # line E
# MARK: each surviving process now processes one data
partition
```

- (a) How many processes (including the original) exist when control reaches MARK?

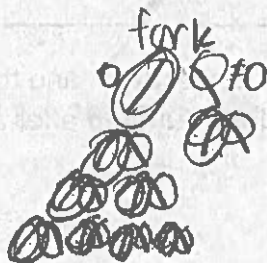
Answer:

18

- (b) How many **child** processes were created in total (i.e., how many times did `os.fork()` return 0)?

Answer:

9



Question 3

[2+1+1 points] An e-commerce platform tracks fulfilled orders in real time. Each warehouse packing station runs as a separate thread and increments a shared counter **orders_fulfilled** after completing each order. The increment is implemented as a non-atomic 3-step read-modify-write:

```

TEMP = orders_fulfilled    # 1. read
TEMP = TEMP + 1            # 2. modify
orders_fulfilled = TEMP    # 3. Write

```

A “lock” with atomic operations **lock()** and **unlock()** is available.

- (a) Where should these operations be placed? Give the line number (L1, etc.) of the line **after** which **lock()** should be inserted, and the line **after** which **unlock()** should be inserted.

```

L1 def station_thread(order_batch):
L2     for order in order_batch:
L3         TEMP = orders_fulfilled
L4         TEMP = TEMP + 1
L5         orders_fulfilled = TEMP

```

Answer – lock() after line number 2

Answer – unlock() after line number 4

- (b) With **5 packing-station threads**, each processing **B = 200 orders**, and the lock correctly placed as in (a), what is the final value of **orders_fulfilled** after all threads finish?

Answer: 2

- (c) Now suppose the locks are removed. The counter currently holds **orders_fulfilled = 75**, and two threads simultaneously execute the 3-step increment (interleaving freely, each completing all three steps once). What is the **smallest** possible value of **orders_fulfilled** after both threads finish?

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Answer: _____

Question 4**[2+1+2 points]** A single-core system runs four processes (time units in ms):P1 takes $t_1 = 25$ *burst*P2 takes $t_2 = 15$ P3 takes $t_3 = 10$ P4 takes $t_4 = 20$ ↓

All arrive at time 0, and there is no overhead for the system switching from one job to another.

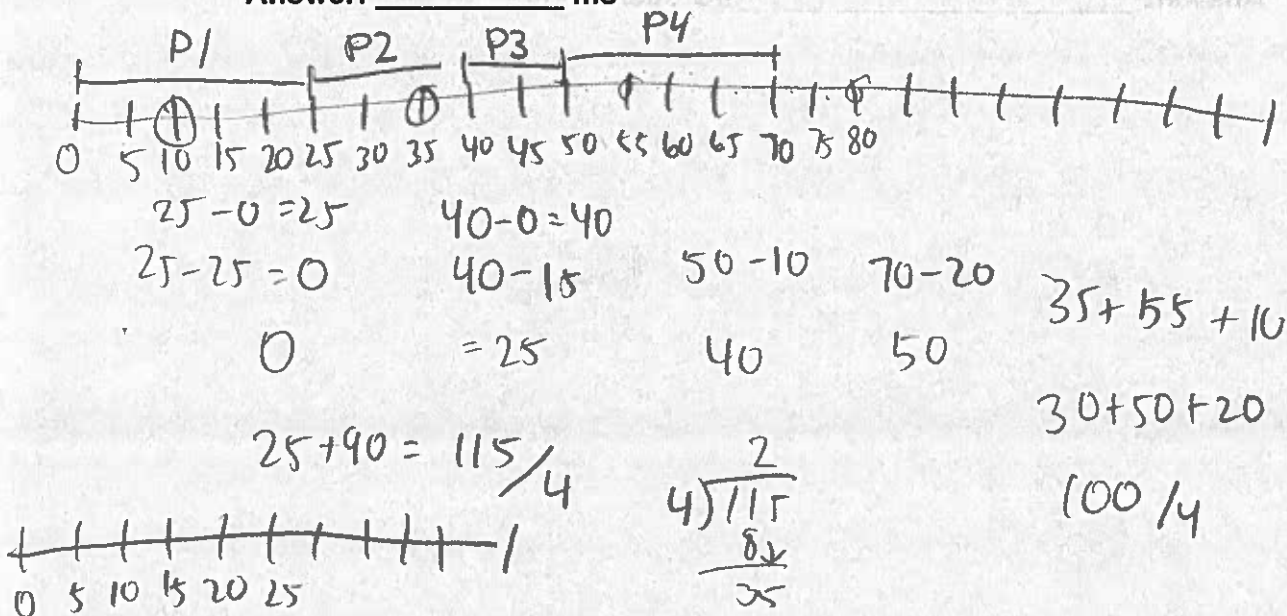
1. Under FIFO scheduling with order (P1, P2, P3, P4), what is the average response (waiting) time? Assume process ID ordering for the same arrival times.

Answer: $\frac{115}{4}$ ms

2. Under SJF (non-preemptive) scheduling, what will be the execution order for P1, P2, P3, P4?

Answer: P3, P2, P4, P1

3. Under SJF (non-preemptive) scheduling, what is the average response (waiting) time?

Answer: $\frac{25}{4}$ ms

Question 5

[2+1+1 points] A data warehousing company uses servers with hard disk drives (HDD) for long-term storage of its customer data.

One of its banking customers undergoes an audit about a past year and needs to pull customer data that total **D = 125 GB** from one of the servers.

The drive where the data are stored has the following specifications:

- Rotation speed $RS = 10000$ rpm (rotations per minute)
- Transfer rate $TR = 250$ MB/s in data chunks of size $d = 0.5$ MB
- For each I/O operation, the seek time is $ST = 4$ ms, and there is an average rotational latency ARL equal to $1/2$ of the rotation period.

For the present exercise, assume $1 \text{ GB} = 1000 \text{ MB}$.

1. What is the time TRC (in ms) needed to transfer a data chunk of size d based on the disk transfer rate?

Answer: 2 ms

2. What is the time T_{IO} (in ms) for one complete I/O?

Answer: 2 ms

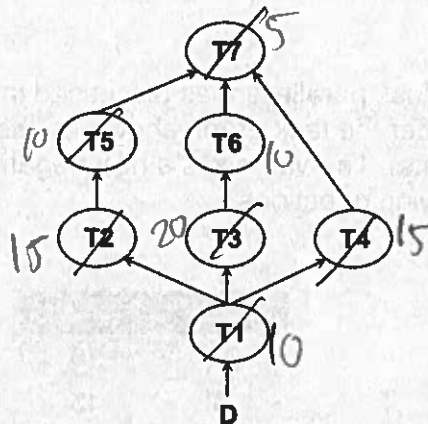
3. What is the number of I/O operations needed to transfer the entire volume of data D?

Answer: 50 I/O operations

Task Parallelism (Task Graphs)

Question 1

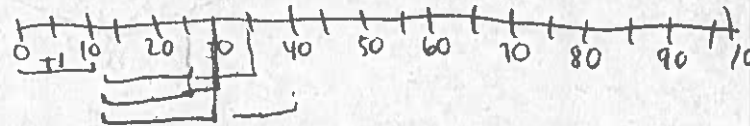
[1+2+3+2+2 points] In this exercise, assume task parallelism as discussed in class (i.e., no preemption or migration of tasks). Consider the task graph shown on the left, the given task lengths in the table (in time units), and answer the following questions.



Task	Length
T1	10
T2	15
T3	20
T4	15
T5	10
T6	10
T7	5

1. What is the largest degree of parallelism, if you were to execute this workload using task parallelism?

Answer: 3



2. Assume you use task parallelism, and you are given as many identical worker nodes as the degree of parallelism in this problem. What is the lowest possible completion time of this workload?

Answer: 40 time units

3. What is the highest possible speedup against running on only one worker node?

Answer: 6 X

$$\frac{1}{8} = \frac{1}{0.333}$$

4. In a task-parallel schedule that offers the highest possible speedup, what is the highest possible idle time of a worker among them all? Assume all workers on, from start to the end of the whole workload.

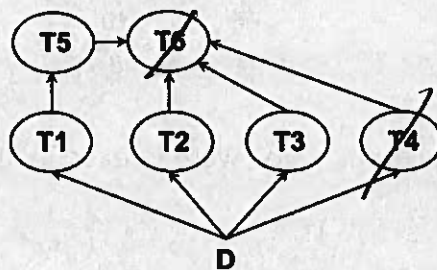
Answer: 30 time units

5. Now suppose you are given only 2 identical worker nodes for task parallelism. What is the lowest possible completion time of this workload now?

Answer: 60 time units

Question 2

[3+2+3 points] In this exercise, assume task parallelism as discussed in class (i.e., no preemption or migration of tasks). Consider the task graph shown on the left, and the given task lengths in the table (in time units). The value x is a non-negative number that can vary as specified in each of the following questions.



Task	Length
T1	10
T2	15
T3	x 25
T4	5
T5	15
T6	10

1. Assume $x=25$. Suppose the workload is executed in a task-parallel manner for the lowest possible completion on as many identical worker nodes as the degree of parallelism in this problem. All workers are on until the last task finishes. What is the total idle time (add across workers)?

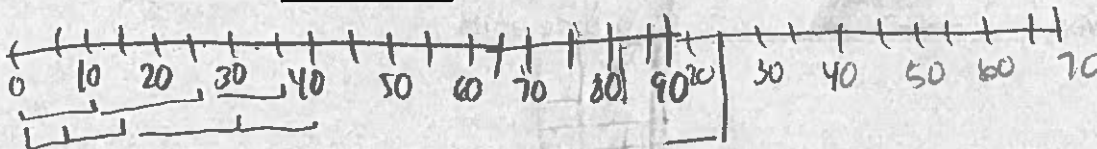
Answer: 215 time units

2. Assume $x=25$. What is the lowest possible completion time of this workload with task-parallelism?

Answer: 25 time units

3. Assume $x=10$. What is the value of the lowest possible completion time with task-parallelism when given only 2 workers?

Answer: 40 time units



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Data Parallelism (SIMD, Amdahl's Law)

Question

[2 points] Preparing the video of your latest Instagram story is a parallelizable task that took 70% of your phone processing resources to complete. What would the speedup be according to Amdahl's law, if your phone could have an infinite amount of processors?

Answer: _____

4 X

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